

Storage System Architectures

Storage systems are categorized by **how data is accessed and managed**. The common types are:

1. Block-Based Storage

- **How it works:**
Data is stored in small, **fixed-size blocks (e.g., 512B, 4KB)**.
Each block is accessed by its address, without any metadata.
- **Used in:** SAN (Storage Area Networks)
- **Examples:** iSCSI, Fibre Channel
- **Best for:** Databases, high-performance applications
- **Needs a file system** to manage files.

2. File-Based Storage

- **How it works:**
Data is stored and accessed **as files with names and directory paths**.
- **Used in:** NAS (Network Attached Storage)
- **Examples:** NFS (Linux/UNIX), SMB/CIFS (Windows)
- **Best for:** File sharing, office documents, media storage
- **Easy to use** with familiar folder structures.

3. Object-Based Storage

- **How it works:**
Data is stored as **objects** – **each object includes the data, metadata, and a unique ID**.
- **Used in:** Cloud storage systems
- **Examples:** **Amazon S3, OpenStack Swift**
- **Best for:** Large unstructured data (videos, images, backups)
- **Highly scalable** and metadata-rich.

4. Unified Storage

- **How it works:**
Combines **block-based and file-based** storage in a single system.
- **Used in:** Mixed environments needing both SAN and NAS.
- **Best for:** Organizations that want flexibility in how data is accessed.
- **Simplifies management** by using a single storage solution.

Summary:

Type	Access Method	Example Use	Suitable For
Block-Based	Blocks	SAN, databases	High performance apps
File-Based	Files/folders	NAS, file servers	File sharing, media, documents
Object-Based	Object ID & metadata	Cloud storage	Backups, media, big data
Unified	Both file & block	Hybrid setups	Versatile, flexible needs

comparison of **Block Storage, File Storage, and Object Storage in Amazon AWS** in a tabular format:

- **EBS (Block)** → Best for **databases**, **OS disks**, and low-latency apps.
- **EFS (File)** → Best for **shared file access across EC2** (like a Linux file server).
- **S3 (Object)** → Best for **cloud-native apps**, **backups, media**, and **web content**.

AWS Storage Types: Comparison Table

Feature / Aspect	Block Storage (EBS)	File Storage (EFS)	Object Storage (S3)
Service Name	Amazon EBS (Elastic Block Store)	Amazon EFS (Elastic File System)	Amazon S3 (Simple Storage Service)
Storage Type	Block-level storage	File-level storage	Object-based storage
Access Method	Attached to a single EC2 instance	Shared via NFS (POSIX-compliant)	Accessed via REST API, SDK, or console
Data Structure	Raw block devices (like a virtual hard disk)	Hierarchical files and directories	Objects with metadata and unique keys
Use Case	Databases, boot volumes, low-latency apps	Shared file systems, web hosting, dev envs	Backups, static websites, media storage, archives
EC2 Attachment	One EC2 instance at a time (can detach/attach)	Multiple EC2 instances simultaneously	Not attached to EC2

Feature / Aspect	Block Storage (EBS)	File Storage (EFS)	Object Storage (S3)
Scalability	Predefined size (can increase)	Automatically scalable	Infinitely scalable
Performance	High IOPS and low latency	Moderate IOPS	Depends on access frequency
Data Access	Via OS-level mount	NFS mount (Linux only)	HTTP/HTTPS API
Backup Support	Snapshots supported	Backup via AWS Backup	Versioning & lifecycle policies
Durability	High (replicated within AZ)	Very High (across multiple AZs)	Very High (11 nines durability)
Cost Model	Pay for provisioned capacity	Pay for used capacity	Pay per GB + API requests

Storage Performance parameters – IOPS, Throughput, Latency

- What is IOPS? – Input Output operations per second e.g. 3000 IOPS
- What is Throughput? – How fast storage can read/write data e.g. 10 MB/s
- What is Latency? – Time delay between request of data and response of data